

Let's Make a Map of India!

A step-by-step tutorial using GeoLibre



What you'll use

- **GeoLibre:** a free, web-based GIS tool, kind of like Google Maps, except you get to build the map yourself
- **india_boundary.geojson:** the outline of the whole country
- **india_state_boundary.geojson:** the outlines of every state inside India

What Is GIS, Anyway?

GIS stands for **Geographic Information System**. That's a big phrase for a simple idea: it's software that lets you put real-world places (countries, rivers, roads, even your school) onto a map, then decorate and study that map however you like.

Today you're going to play the role of a mapmaker, called a **cartographer**. By the end of this tutorial you'll have built the map on the cover: India, photographed from space, with a bold black outline around the country and every state, plus a title, a north arrow, and a scale bar, just like a map you'd see in an atlas.

GIS WORD OF THE DAY

Layer: In GIS, a map is built like a sandwich. Each slice is called a *layer*: one layer might be a photo of the ground, another might be country borders, another might be city names. You stack layers on top of each other to build the final map.

FUN FACT

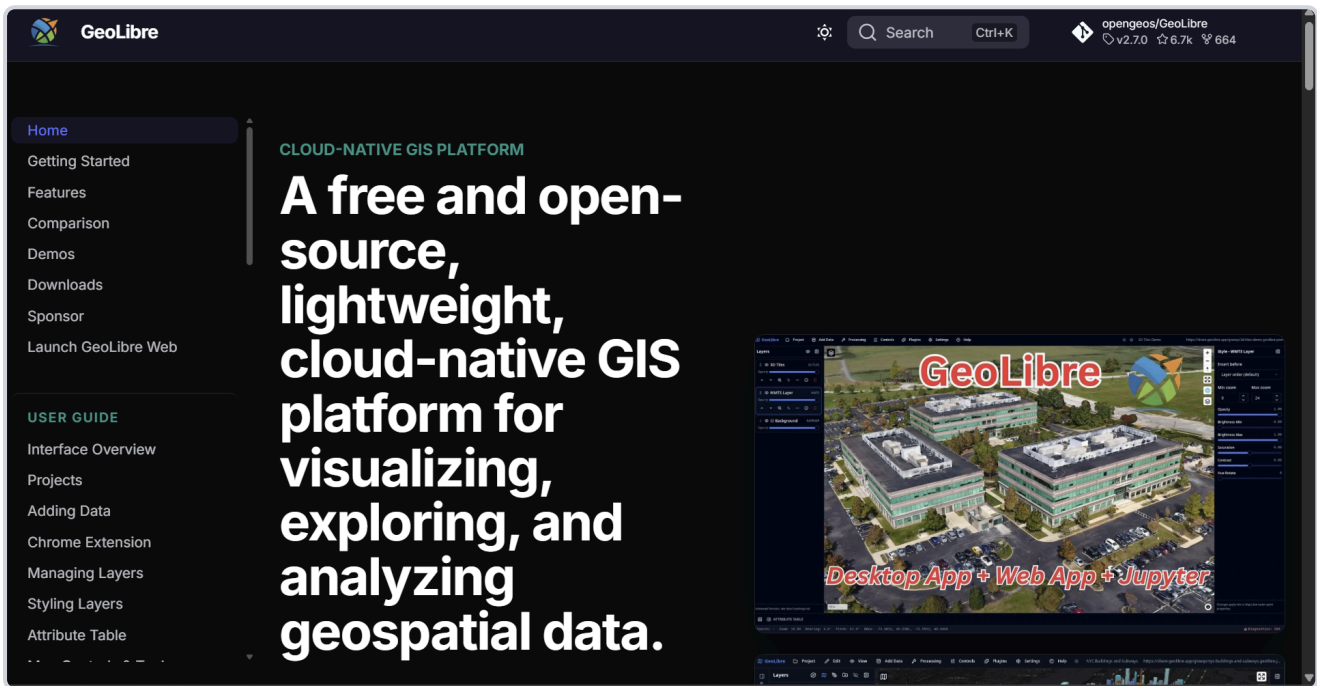
India is the 7th largest country in the world by area, and it shares borders with six other countries: Pakistan, China, Nepal, Bhutan, Bangladesh, and Myanmar.

Here's the plan for today, in seven steps:

1. Open GeoLibre
2. Add the country outline layer
3. Add the state outline layer
4. Style both layers (make them see-through with a dark border)
5. Switch the background to a satellite photo
6. Add a title, north arrow, and scale bar
7. Export your finished map

1 Open GeoLibre

1. Open your web browser and go to the GeoLibre website.
2. You'll land on the homepage shown below. Look for the button called **"Launch GeoLibre Web"** in the left-hand menu, and click it.
3. A blank map of the whole world will open. That's your empty canvas, ready for you to build on.



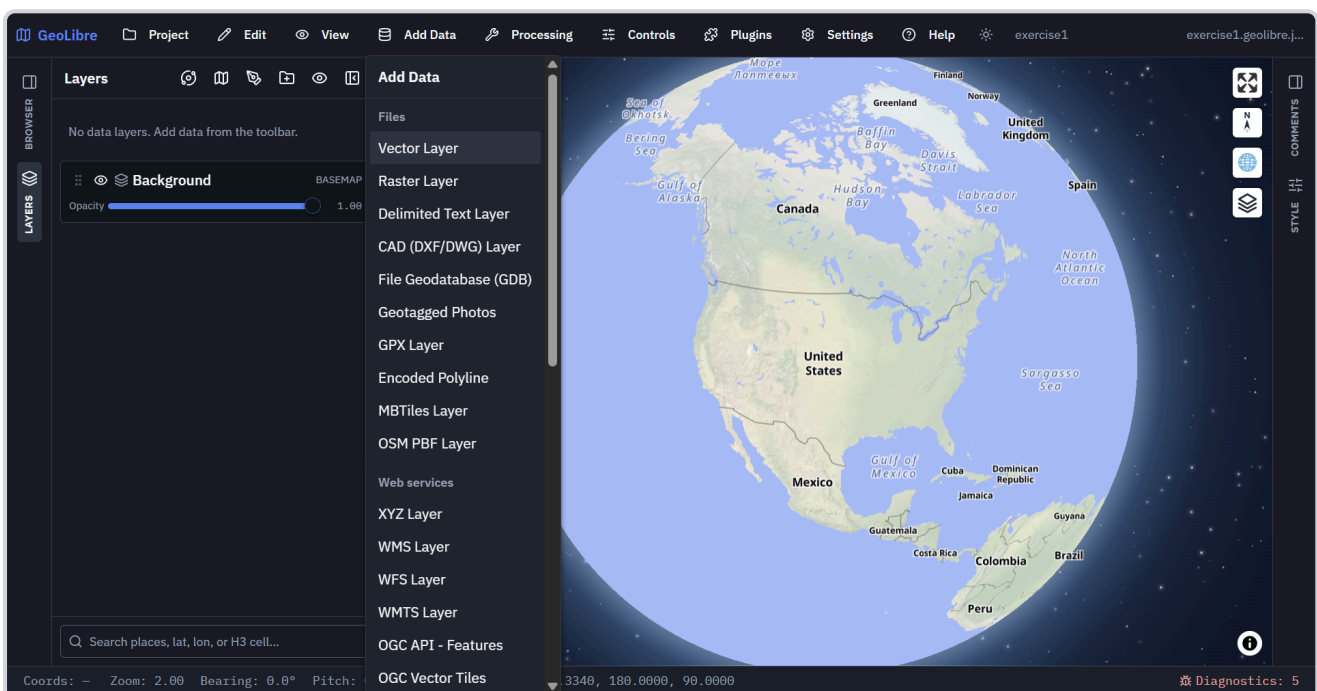
The GeoLibre homepage. It's free and open-source, so anyone can use it and see how it was built.

TIP

You don't need to install anything. GeoLibre runs entirely in your browser, so it works the same on Windows, Mac, or a school Chromebook.

Now let's put our first layer onto the map: the outline of India.

1. Along the top menu bar, click **Add Data**.
2. A panel slides open with a list of ways to bring data into your map. Under the **Files** section, click **Vector Layer**.
3. Browse to your data folder and choose **india_boundary.geojson**, then confirm. The outline of India will appear on your map, listed as a new layer named *india_boundary* in the Layers panel on the left.



The Add Data panel, open and ready. "Vector Layer" is the option you want for a .geojson file.

GIS WORD OF THE DAY

Vector data: shapes made of points, lines, and polygons (like a country border) instead of a grid of colored dots (that's called *raster data*, and it's what a satellite photo is made of).

GeoJSON is just a common file format for storing vector shapes, written in plain text.

3 Add the State Lines

One layer down, one to go! Repeat the exact same process to add the internal state borders.

1. Click **Add Data** again.
2. Choose **Vector Layer** again.
3. This time pick **india_state_boundary.geojson**.
4. Check your Layers panel: you should now see two layers listed, *india_boundary* and *india_state_boundary*, stacked on top of the world basemap.

FUN FACT

India is made up of 28 states and 8 union territories, so that's a lot of borders. The state boundary file you just loaded contains every single one of those lines.

TIP

Order matters in the Layers panel. Since you added the state boundary layer after the country boundary layer, it should sit above it in the list, which is exactly what we want.

4 Style Your Map

Right now both layers are probably filled in with a solid color, which hides everything underneath. Let's turn them into simple outlines instead. That way, once we add a satellite photo underneath in the next step, you'll be able to see the real ground through the shapes.

1. Click on the **india_boundary** layer to select it, then open its **Style** settings.
2. Set the **Fill Color** to **Transparent** (see-through, no color at all).
3. Set the **Line Color** to a dark, almost-black color, and the **Line Width** to **2**.
4. Repeat for the **india_state_boundary** layer, but this time set the **Line Width** to **1**, a little thinner, so the country outline stands out more than the state lines.

Layer	Fill Color	Line Color	Line Width
india_boundary	Transparent	 #191c25	2
india_state_boundary	Transparent	 #191c25	1

GIS WORD OF THE DAY

Style: the colors, line widths, and patterns used to draw a layer. The data (where the border actually is on Earth) never changes, but the style controls how it looks on your screen.

5 Switch to Satellite View

Your map still has a plain background. Let's replace it with a real photo of Earth taken from space.

1. Find the **basemap** button (it looks like a small globe icon on the map, or check the **Add Data** → **Web services** options).
2. From the list of basemaps, choose **Google Satellite**.
3. The plain background turns into an actual satellite photo of the Earth's surface: mountains, deserts, forests, and oceans, all real.
4. Because you made your two layers transparent in the last step, the black country and state outlines now sit neatly on top of the photo.

GIS WORD OF THE DAY

Basemap: the background layer of a map, usually a photo or a simple street map, that gives you context for everything drawn on top of it.

FUN FACT

Look closely at the northern edge of India on the satellite photo. Those white, wrinkled patches are the Himalayas, home to Mount Everest and the highest mountains on Earth.



Your map on a satellite basemap. The black outline sits right on top of the real Earth.

A real map always tells you three things: what it's a map of, which way is north, and how big things really are. Let's add all three.

1. Open the **print layout** (sometimes called the map layout or export panel) from the **Controls** menu.
2. Type "**Map of India**" into the Title field, and turn on **Show Title**.
3. Turn on **Show North Arrow**. This adds an arrow pointing "up" toward true north, so anyone reading your map knows which direction they're facing.
4. Turn on **Show Scale Bar**. This adds a small ruler in a corner of the map showing how many real kilometers fit into a given length on the page.
5. Set the paper size to **A4** and the orientation to **Portrait** (taller than it is wide), which fits India's shape nicely.

GIS WORD OF THE DAY

Scale bar: a small ruler on a map. If it says "500 km" next to a short black-and-white bar, that means the real distance covered by that bar, on the actual ground, is 500 kilometers.

7

Export and Admire!

1. Once your title, north arrow, and scale bar all look right, use the **Export** (or Print/Download) button to save your map as an image.
2. Open the image and compare it to the cover of this tutorial. Do the outlines match up? Can you spot the Himalayas, the coastline, and Sri Lanka floating just off the southern tip of India?

Congratulations, you just made a real map from scratch, just like the one on the cover!

What You Learned

How **layers** stack like a sandwich to build one map

Loading shapes from GeoJSON files as **vector data**

Setting fill color, line color, and line width when **styling** a layer

Swapping in a satellite photo as your **basemap**

Adding a title, north arrow, and scale bar as **map elements**

Exporting your project as a shareable image

Try It Yourself

- Change the outline color from black to bright red or blue. Does the map look different to you?
- Turn off the *india_state_boundary* layer's visibility. How does the map change with just the country outline showing?
- Write your own title, like "My Map of India" or today's date, and export a second version.
- Try switching the basemap to a plain street map instead of satellite. Which style do you like better, and why?